

ASPECTS OF NON-SINGULAR ERGODIC THEORY

KATHMANDU, NOVEMBER 2014.

JON AARONSON'S LECTURE NOTES

Lecture # 1 11/11/2014.

INTRODUCTION

Standard spaces.

A *standard measurable space* $(X, \mathcal{B})$ is a complete, separable metric space X (aka *Polish*) equipped with its Borel sets $\mathcal{B} = \sigma(\{\text{open sets}\})$.

For example, $X = [0, 1]$, $\Omega := \{0, 1\}^{\mathbb{N}}$, $\mathbb{R}^d$, $C([0, 1])$...

Kuratowski's isomorphism theorem

Let X be an uncountable Polish space, then $X \cong \Omega$ i.e. $\exists$ a bimeasurable bijection $\pi : X \rightarrow \Omega$.

A standard measure space $(X, \mathcal{B}, m)$ is a standard measurable space $(X, \mathcal{B})$ equipped with a σ -finite, non-atomic measure $m : \mathcal{B} \rightarrow [0, \infty]$.

We distinguish the cases where m is a probability, and $m(X) = \infty$. Using Kuratowski's theorem, it is not hard to see that

- a standard probability space $(\Omega, \mathcal{F}, P) \cong ([0, 1], \mathcal{B}([0, 1]), \text{Leb})$ &
- a standard, infinite measure space $(X, \mathcal{B}, m) \cong (\mathbb{R}, \mathcal{B}(\mathbb{R}), \text{Leb})$.

Equivalent measures and Radon-Nikodym derivatives.

Let $(X, \mathcal{B})$ be a measurable space and let $m, \mu : \mathcal{B} \rightarrow [0, \infty)$ be finite measures. We say that μ is *absolutely continuous wrt* m ($\mu \ll m$) if $m(A) = 0 \Rightarrow \mu(A) = 0$ and we say that μ & m are *equivalent* ($\mu \sim m$) if $\mu \ll m$ & $m \ll \mu$.

For example if $f : X \rightarrow \mathbb{R}$, $f \geq 0$ is m -integrable and $\mu(A) := \int_A dm$, then $\mu \ll m$ and $\mu \sim m$ in case $f > 0$ a.s..

Radon Nikodym Theorem If $\mu \ll m$ then $\exists f \in L^1(m)_+$ so that $\mu(A) := \int_A dm$.

Non-singular transformations. Let $(X, \mathcal{B}, m)$ be a standard measure space.

A **null preserving transformation** (NPT) of X is only defined modulo nullsets, and is a map $T : X_0 \rightarrow X_0$ (where $X_0 \subset X$ has full measure), which is measurable and has the *null preserving property* that for $A \in \mathcal{B}$, $m(T^{-1}A) = 0$ implies that $m(A) = 0$.

A *non-singular transformation* (NST) is a NPT $(X, \mathcal{B}, m, T)$ with the stronger property that for $A \in \mathcal{B}$, $m(T^{-1}A) = 0$ iff $m(A) = 0$.

A *measure preserving transformation* (MPT) is a NST $(X, \mathcal{B}, m, T)$ with the additional property that $m(T^{-1}A) = m(A) \forall A \in \mathcal{B}$.

We'll call a nonsingular transformation **NS-invertible** if the associated map is invertible with a nonsingular inverse.

Equivalent invariant measures. If T is a non-singular transformation of a σ -finite measure space $(X, \mathcal{B}, m)$, and p is another measure on $(X, \mathcal{B})$ *equivalent* to m (denoted $p \sim m$ and meaning that p and m have the same nullsets), then T is a non-singular transformation of $(X, \mathcal{B}, p)$.

Thus, a non-singular transformation of a σ -finite measure space is actually a non-singular transformation of a probability space.

The first question about a NST $(X, \mathcal{B}, p, T)$ is whether it was obtained from a measure preserving transformation in this way, or, slightly more generally:

$\exists ?$ a σ -finite absolutely continuous invariant measure (a.c.i.m., i.e. $m \ll p$, with $m \circ T^{-1} = m$).

RADON NIKODYM DERIVATIVES

Let $(X, \mathcal{B}, m, T)$ be an invertible NST of the probability space $(X, \mathcal{B}, m)$. The measures m & $m \circ T$ are equivalent (i.e. $m \circ T \ll m$ & $m \ll m \circ T$), written $m \circ T \sim m$. By the Radon Nikodym theorem, $\exists ! T' \in L^1$, $T' > 0$ a.e., so that

$$m(TA) = \int_A T' dm \quad \forall A \in \mathcal{B}.$$

The function $T' = \frac{dm \circ T}{dm}$ is called the *RN derivative of T with respect to m* .

Transfer Operator.

Let $(X, \mathcal{B}, m, T)$ be a null-preserving transformation, then $\|f \circ T\|_\infty \leq \|f\|_\infty \forall f \in L^\infty(m)$ and $T : L^\infty(m) \rightarrow L^\infty(m)$ where $Tf := f \circ T$.

There is an operator known as the *transfer operator* $\widehat{T} : L^\infty(m) \rightarrow L^\infty(m)$ so that $\widehat{T}^* = T$ i.e.:

$$\int_X \widehat{T}f \cdot g dm = \int_X f \cdot Tg dm \quad \forall f \in L^1(m), g \in L^\infty(m).$$

This is given by $\widehat{T}f := \frac{d\nu_f \circ T^{-1}}{dm}$ where $\nu_f(A) := \int_X f dm$ (!).

EXAMPLES

The adding machine. Let $\Omega = \{0, 1\}^\mathbb{N}$, and $\mathcal{F}$ be the σ -algebra generated by cylinders. Define the *adding machine* $\tau : \Omega \rightarrow \Omega$ by $\tau(\bar{1}) := (\bar{0})$ where $(\bar{a})_k = a \quad \forall k \geq 1$; and

$$\tau(1, \dots, 1, 0, \omega_{\ell+1}, \omega_{\ell+2}, \dots) = (0, \dots, 0, 1, \omega_{\ell+1}, \omega_{\ell+2}, \dots)$$

for $\omega \in \Omega \setminus \{(\bar{1})\}$ where $\ell(\omega) := \min\{n \geq 1 : \omega_n = 0\}$.

The reason for the name "adding machine" is that

$$\sum_{k=1}^{\infty} 2^{k-1} (\tau^n \bar{0})_k = n \quad \forall n \geq 1.$$

We'll consider the adding machine with respect to various probabilities on Ω .

¶ For $p \in (0, 1)$, define a probability μ_p on Ω by

$$\mu_p([\epsilon_1, \dots, \epsilon_n]) = \prod_{k=1}^n p(\epsilon_k)$$

where $p(0) = 1 - p$ and $p(1) = p$.

1.3 Proposition

τ is an invertible, nonsingular transformation of $(\Omega, \mathcal{F}, \mu_p)$ with

$$\frac{d\mu_p \circ \tau}{d\mu_p} = \left(\frac{1-p}{p} \right)^{\ell-2}.$$

Proof

We show that $\mu_p \circ \tau \sim \mu_p$ and calculate $\frac{d\mu_p \circ \tau}{d\mu_p}$. We show that for any set $A \in \mathcal{F}$,

$$\mu_p(\tau A) = \int_A \left(\frac{1-p}{p} \right)^{\ell-2} d\mu_p.$$

Consider first a cylinder set $A \subset [\ell = k] \quad (k \geq 1)$

$$A = [\underbrace{1, \dots, 1}_{k-1 \text{ times}}, 0, a_1, \dots, a_n],$$

then

$$\tau A = [\underbrace{0, \dots, 0}_{k-1 \text{ times}}, 1, a_1, \dots, a_n],$$

and

$$\begin{aligned} (\spadesuit) \quad \mu_p(\tau A) &= \mu_p([\underbrace{0, \dots, 0}_{k-1 \text{ times}}, 1]) \mu_p([a_1, \dots, a_n]) \\ &= \left(\frac{1-p}{p} \right)^{k-2} \mu_p(A) \\ &= \int_A \left(\frac{1-p}{p} \right)^{\ell-2} d\mu_p. \end{aligned}$$

Let

$$\mathcal{C} := \{A \in \mathcal{F} : (\spadesuit) \text{ holds}\}.$$

As above, $\mathcal{C} \supset \{\text{cylinders}\}$.

Since any finite union of cylinders is also a finite union of disjoint cylinders, $\mathcal{C} \subset \mathcal{A}$, the algebra of finite unions of cylinders.

By σ -additivity of μ_p , $\mathcal{C}$ is a monotone class, and by the monotone class theorem, $\mathcal{C} \supseteq \sigma(\mathcal{A}) = \mathcal{B}$. $\square$

Note that $\mu_{\frac{1}{2}} \circ \tau = \mu_{\frac{1}{2}}$.

Rank one constructions.

This method constructs a $T \in \text{MPT}(X, \mathcal{B}, m)$ where $X = (0, S_T)$ is an interval, m is Lebesgue measure and where T is an invertible *piecewise translation* that is there are intervals $\{I_n : n \geq 1\}$ and numbers $a_n \in \mathbb{R}$ ($n \geq 1$) so that mod m :

$$X = \bigcup_{n=1}^{\infty} I_n = \bigcup_{n=1}^{\infty} (a_n + I_n) \quad \& \quad T(x) = x + a_n \text{ for } x \in I_n.$$

The **rank one transformation** $(X, \mathcal{B}, m, T)$ is an invertible piecewise translation of an interval $J_T = (0, S_T)$ where $S_T \in (0, \infty]$ which is defined as the “limit of a refining sequence of Rokhlin towers”.

- A *Rokhlin tower* is a finite sequence of disjoint intervals $\tau = (I_1, I_2, \dots, I_n)$ of equal lengths; considered equipped with the translations $I_j \rightarrow I_{j+1}$ ($1 \leq j \leq n-1$). It is thus a piecewise translation

$$T_\tau : \text{Dom } T_\tau = \bigcup_{j=1}^{n-1} I_j \rightarrow \bigcup_{j=2}^n I_j$$

being defined everywhere on $\bigcup_{j=1}^n I_j$ except the last interval I_n .

- We'll say that the Rokhlin tower $\theta = (J_1, \dots, J_\ell)$ *refines* the Rokhlin tower $\tau = (I_1, I_2, \dots, I_n)$ (written $\theta > \tau$) if

$$\bigcup_{j=1}^n I_j \subset \bigcup_{k=1}^{\ell} J_k \text{ \& } I_j = \bigcup_{1 \leq k \leq \ell, J_k \subset I_j} J_k.$$

This entails (!) $\bigcup_{j=1}^{n-1} I_j \subset \bigcup_{k=1}^{\ell-1} J_k$, whence $T_\theta|_{\bigcup_{j=1}^{n-1} I_j} \equiv T_\tau$.

Definition.

Let $c_n \in \mathbb{N}$, $c_n \geq 2$ ($n \geq 1$) and let $S_{n,k} \geq 0$, ($n \geq 1$, $1 \leq k \leq c_n$). The *rank one transformation with construction data*

$$\{(c_n; S_{n,1}, \dots, S_{n,c_n}) : n \geq 1\}$$

is an invertible piecewise translation of the interval $J_T = (0, S_T)$ where

$$S_T := 1 + \sum_{n \geq 1} \frac{1}{c_1 \cdots c_n} \sum_{k=1}^{c_n} S_{n,k} \leq \infty.$$

To obtain T , we define a refining sequence $(\tau_n)_{n \geq 1}$ of Rokhlin towers where $\tau_1 = [0, 1]$ and τ_{n+1} is constructed from τ_n by

- cutting τ_n into c_n columns of equal width,
- putting $S_{n,k}$ spacer intervals (of the same width) above the k^{th} column ($1 \leq k \leq c_n$);
- and stacking.

Evidently $\tau_{n+1} > \tau_n$. Let X be the increasing union of the intervals in the towers τ_n .

The sum of the lengths of the last intervals of the towers is $\sum_{n=1}^{\infty} \frac{1}{c_1 \cdots c_n} < \infty$ and so for a.e. $x \in X$, $\exists n \leq 1$ so that $x \in \text{Dom } T_{\tau_k} \ \forall k \geq n$ and $T(x) := T_{\tau_k}(x) \ \forall k \geq n$.

The length of X is 1 plus the total length of all the spacer intervals added in the construction i.e. S_T .

Kakutani skyscrapers.

Suppose that $(\Omega, \mathcal{F}, \mu, S)$ is a PPT of the probability space $((\Omega, \mathcal{F}, \mu)$ and that $\varphi : \Omega \rightarrow \mathbb{N}$ is measurable. The *Kakutani skyscraper* over S with *height function* φ is the transformation T of the σ -finite measure space $(X, \mathcal{B}, m)$ defined as follows.

$$X = \{(x, n) : x \in \Omega, 1 \leq n \leq \varphi(x)\},$$

$$\mathcal{B} = \sigma\{A \times \{n\} : n \in \mathbb{N}, A \in \mathcal{F} \cap [\varphi \geq n]\}, \ m(A \times \{n\}) = \mu(A),$$

and

$$T(x, n) = \begin{cases} (Sx, \varphi(x)) & \text{if } n = \varphi(x), \\ (x, n+1) & \text{if } 1 \leq n \leq \varphi(x) - 1. \end{cases}$$

Evidently T is a MPT with

$$m(X) = \int_{\Omega} \varphi d\mu.$$

Moreover,

- $\bigcup_{n \geq 1} T^{-n}(\Omega \times \{1\}) = X$;
- For $x \in \Omega$, let $\varphi_N(x) := \sum_{k=0}^{N-1} \varphi(S^k x)$, then $T^{\varphi_N(x)}(x, 1) = (S^N x, 1)$ and

$$\{n \geq 1 : T^n(x, 1) \in \Omega \times \{1\}\} = \{\varphi_N(x) : N \geq 1\}.$$

Bernoulli shift.

The (two sided) *Bernoulli shift* is defined by $X = \mathbb{R}^{\mathbb{Z}}$, $\mathcal{B}(X)$ the σ -algebra generated by *cylinder sets* of form

$$[A_1, \dots, A_n]_k := \{\underline{x} \in X : x_{j+k} \in A_j, 1 \leq j \leq n\}$$

where $A_1, \dots, A_n \in \mathcal{B}(\mathbb{R})$. The shift $S : X \rightarrow X$ is defined by $(Sx)_n = x_{n+1}$.

Let $p : \mathcal{B}(\mathbb{R}) \rightarrow [0, 1]$ be a probability, and let $\mu_p : \mathcal{B}(X) \rightarrow [0, 1]$ be that probability measure so that

$$\mu_p([A_1, \dots, A_n]_k) = \prod_{k=1}^n p(A_k) \quad (A_1, \dots, A_n \in \mathcal{B}(\mathbb{R})).$$

Evidently (!), the two sided Bernoulli shift is measure preserving.

Interval maps.

Let $I \subseteq \mathbb{R}$ be an interval, let m be Lebesgue measure on I , and α be a collection of disjoint open subintervals of I such that

$$m(I \setminus U_{\alpha}) = 0 \text{ where } U_{\alpha} = \bigcup_{a \in \alpha} a.$$

For $r \geq 1$, a C^r *interval map* with *basic partition* α is a map $T : I \rightarrow I$ such that

for each $a \in \alpha$, $T|_a$ extends to a C^r diffeomorphism $T : \bar{a} \rightarrow T(\bar{a})$.

The C^r interval map is called *piecewise onto* if $T(a) = I \forall a \in \alpha$.

Transfer operator of an interval map.

Let $T : I \rightarrow I$ be a C^r , interval map with basic partition α . For $a \in \alpha$, let $v_a : I \rightarrow a$ be the inverse of $T : a \rightarrow I$ (a C^r diffeomorphism). It follows from an integration variable-change argument that with respect to m :

$$\widehat{T}f = \sum_{a \in \alpha} 1_{T(a)} v'_a f \circ v_a$$

Here $v'_a := \frac{dm \circ v_a}{dm} = \left| \frac{dv_a}{dx} \right|$.

For each $n \geq 2$, $T^n : I \rightarrow I$ is a, C^r interval map with basic partition

$$\alpha_n := \{[a_0, a_1, \dots, a_{n-1}] := \bigcap_{i=0}^{n-1} T^{-k} a_i : a_0, a_1, \dots, a_{n-1} \in \alpha\}.$$

For $a \in \alpha_n$, let $v_a : I \rightarrow a$ be the inverse of $T^n : a \rightarrow I$, then

$$\widehat{T}^n f = \widehat{T^n} f = \sum_{a \in \alpha_n} 1_{T^n(a)} v'_a f \circ v_a.$$

Exercise 4.

Show that if (I, T, α) is a piecewise onto, piecewise linear interval map (i.e. $T : a \rightarrow Ta$ is affine $\forall a \in \alpha$) with $\#\alpha \geq 2$, then $m \circ T^{-1} = m$ and

$$m\left(\bigcap_{k=0}^N T^{-k} a_k\right) = \prod_{k=0}^N m(a_k) \quad \forall N \geq 1, a_0, a_1, \dots, a_N \in \alpha.$$

Boole transformations & inner functions.

A *Boole transformation* is a map $T : \mathbb{R} \rightarrow \mathbb{R}$ of form

$$T(x) = \alpha x + \beta + \sum_{k=1}^N \frac{p_k}{t_k - x}$$

where $\alpha \geq 0$, $p_1, \dots, p_N > 0$ & $\beta, t_1, \dots, t_N \in \mathbb{R}$.

A Boole transformation T is an *inner function* of the upper half plane $\mathbb{R}^{2+} := \{\omega \in \mathbb{C} : \text{Im} \omega > 0\}$ i.e. an analytic endomorphism of $\mathbb{R}^{2+}$ which preserves $\mathbb{R}$.

The general form of an inner function T of $\mathbb{R}^{2+}$ is given by:

$$(\clubsuit) \quad T(\omega) = \alpha \omega + \beta + \int_{\mathbb{R}} \frac{1 + t\omega}{t - \omega} d\mu(t)$$

where $\alpha \geq 0$, $\beta \in \mathbb{R}$ and μ is a finite, Lebesgue-singular, measure on $\mathbb{R}$.

If $\omega \in \mathbb{R}^{2+}$ the upper half plane, and $\omega = a + ib$, $a, b \in \mathbb{R}$, $b > 0$ then

$$\text{Im} \frac{1}{x - \omega} = \frac{b}{(x - a)^2 + b^2} = \pi \varphi_\omega(x)$$

where φ_ω is the well known *Cauchy density*.

These are the densities of the **Poisson** or **harmonic** measures on $\mathbb{R}^{2+}$:

If $\phi : \mathbb{R}^{2+} \rightarrow \mathbb{C}$ is bounded, analytic on $\mathbb{R}^{2+}$ and then for a.e. $t \in \mathbb{R}$, $\exists \lim_{y \rightarrow 0+} \phi(t + iy) =: \phi^*(t)$ and

$$(\mathfrak{A}) \quad \phi(\omega) = \int_{\mathbb{R}} \phi^*(t) dP_{\omega}(t) \quad (\omega \in \mathbb{R}^{2+})$$

where $dP_{\omega}(t) = \varphi_{\omega}(t) dt$.

2.2 Boole's Formula *Let T be an inner function, then $(\mathbb{R}, \mathcal{B}, m, T)$ is non-singular and*

$$(\mathfrak{B}) \quad \widehat{T} \varphi_{\omega} = \varphi_{T(\omega)} \quad \forall \omega \in \mathbb{R}^{2+}.$$

Proof (G.Letac) It suffices to show that $P_{\omega} \circ T^{-1} = P_{T(\omega)}$.

The Fourier transform of P_{ω} is given by

$$\widehat{P_{\omega}}(t) := \int_{\mathbb{R}} e^{itx} dP_{\omega}(x) = e^{it\omega} \quad (t \geq 0).$$

For $t > 0$, $\phi_t(\omega) = e^{it\omega}$ is a bounded analytic functions on $\mathbb{R}^{2+}$ with $\phi_t^*(x) = e^{itx}$ on $\mathbb{R}$. By $(\mathfrak{A})$,

$$\widehat{P_{\omega} \circ T^{-1}}(t) = \int_{\mathbb{R}} e^{itT(x)} dP_{\omega}(x) = e^{itT(\omega)} = \widehat{P_{T(\omega)}}(t),$$

whence $(\mathfrak{B})$. $\square$

Lecture # 2 12/11/2014.**INVARIANT PROBABILITIES**

Let $T : \mathbb{R}^{2+} \rightarrow \mathbb{R}^{2+}$ be a non-Möbius inner function. By the Denjoy-Woelf theorem . $\exists \beta \in \overline{\mathbb{R}^{2+}} \cup \{\infty\}$ so that $T^n(\omega) \xrightarrow[n \rightarrow \infty]{} \beta \forall \omega \in \mathbb{R}^{2+}$.

The point β is aka the **Denjoy-Wolff point** of T .

If $\beta \in \mathbb{R}^{2+}$, then $T(\beta) = \beta$ and $P_\beta \circ T^{-1} = P_\beta$.

Otherwise (!) $\widehat{T}^n \varphi_\omega \xrightarrow[n \rightarrow \infty]{} 0 \forall \omega \in \mathbb{R}^{2+}$ and $\nexists$ an a.c.i.p. for T .

Proposition 2.1

If T is an inner function with $\alpha > 0$ in $(\mathbb{H})$, then $m \circ T^{-1} = \frac{1}{\alpha} \cdot m$.

Vague sketch of proof that $\widehat{T}\mathbb{1} = \frac{1}{\alpha}\mathbb{1}$

- $\pi b \varphi_{ib} \xrightarrow[b \rightarrow \infty]{} 1$ unifomly on bounded subsets of $\mathbb{R}$;
- if $T(ib) = u(b) + iv(b)$, then $\frac{v(b)}{b} \xrightarrow[b \rightarrow \infty]{} \alpha$ & $\frac{u(b)}{b} \xrightarrow[b \rightarrow \infty]{} 0$.
- $\widehat{T}\mathbb{1} \xleftarrow[b \rightarrow \infty]{} \pi b \widehat{T} \varphi_{ib} = \pi b \varphi_{T(ib)} \xrightarrow[b \rightarrow \infty]{} \frac{1}{\alpha} \mathbb{1}. \quad \square$

Exercise 5: Boole & Glaisher transformations.

For $\alpha, \beta > 0$ define $T = T_{\alpha, \beta} : \mathbb{R} \rightarrow \mathbb{R}$ by $T(x) := \alpha x - \frac{\beta}{x}$.

(a) Show that if $\alpha + \beta = 1$, then $\widehat{T}\varphi_i = \varphi_i$ and T has an absolutely continuous, invariant probability (a.c.i.p.).

Consider the Glaisher transformations $T : \mathbb{R} \rightarrow \mathbb{R}$ of form

$$T_{a,b}x := ax + b \tan x \quad (a, b \geq 0, a + b > 0).$$

(b) Give conditions on a, b so that $T_{a,b}$ has an absolutely continuous invariant probability.

(c) Show that $T_{1,b}$ preserves Lebesgue measure.

(d) Show that $T_{0,1}x = \tan x$ preserves the measure $d\mu_0(x) := \frac{dx}{x^2}$.

Hint: $S := \pi \circ T_{0,1} \circ \pi^{-1}$ preserves Lebesgue measure where $\pi(x) := \frac{-1}{x}$.

Existence of invariant probabilities.

It is not always possible to identify absolutely continuous invariant probabilities by explicit formulae as above.

To help remedy this situation, there are many conditions for existence of such. Some conditions for existence of absolutely continuous invariant probabilities depend on the following:

2.2 Invariant probability proposition *Let T be a nonsingular transformation of $(X, \mathcal{B}, m)$.*

If $\exists f \in L^1(m)$, $f \geq 0$, $\int_X f dm > 0$ such that $\{\frac{1}{n} \sum_{k=1}^n \widehat{T}^k f : n \geq 1\}$ is a uniformly integrable family, then $\exists$ a T -invariant probability $P \ll m$.

Proof

As is well known, a uniformly integrable family is weakly sequentially compact in $L^1(m)$. Any weak limit point $h \in L^1(m)$ of $\{\frac{1}{n} \sum_{k=1}^n \widehat{T}^k f : n \geq 1\}$ (where $f \in L^1(m)$, $f \geq 0$, $\int_X f dm = 1$) is the density of a T -invariant probability. $\square$

Expanding, C^2 interval maps. The piecewise onto, C^2 interval map T is called *uniformly expanding* if

$$(UE) \quad \exists \lambda > 1 \ni |T'x| \geq \lambda \quad \forall x \in I;$$

and satisfies *Adler's condition* of "bounded distortion" if

$$(A) \quad \exists M > 1 \ni \frac{|T''x|}{|T'x|^2} \leq M \quad \forall x \in I.$$

This is automatic in case α is finite.

2.3 Renyi's theorem

Each piecewise onto, C^2 , uniformly expanding interval map T satisfying Adler's condition has an a.c.i.p.

Proof in case $m(I) = 1$

Note first that Adler's condition holds uniformly for all powers of T :

$$(\clubsuit) \quad \frac{|T^{n''}x|}{|T^{n'}x|^2} \leq K = \frac{\lambda M}{\lambda - 1} \quad \forall x \in \bigcap_{k=0}^{n-1} T^{-k}U_\alpha, \quad n \geq 1.$$

Next, using that for $a \in \alpha_n$, $v'_a = \frac{-1}{T^{n'}ov_a}$, we see that $(\clubsuit)$ entails that $\forall n \geq 1, a \in \alpha_n$,

$$(\spadesuit) \quad |v''_a| \leq K|v'_a| \quad \text{a.e. on } I$$

where $K := \frac{\lambda M}{\lambda - 1}$; whence $(!!)$ v'_a does not change sign on I and

$$\frac{d}{dx} \log |v'_a| = \frac{v''_a}{v'_a}$$

with the consequence that

$$|v'_a| = e^{\pm K} m(a) \quad \forall a \in \alpha_n, \quad n \geq 1.$$

Thus, for $n \geq 1$,

$$\widehat{T}^n 1 = \sum_{a \in \alpha_n} |v'_a| = e^{\pm K},$$

$$\begin{aligned}
(\widehat{T}^n 1)' &= \sum_{a \in \alpha_n} (|v'_a|)' \\
&\leq K \sum_{a \in \alpha_n} |v'_a| \\
&\leq K e^K.
\end{aligned}$$

By the Arzela-Ascoli theorem, $\exists n_k \rightarrow \infty$ and $H : I \rightarrow \mathbb{R}_+$ so that

$$\frac{1}{n_k} \sum_{j=0}^{n_k-1} \widehat{T}^j 1 \xrightarrow[k \rightarrow \infty]{} H \quad \text{uniformly on } I;$$

whence H is Lipschitz continuous and the density of an a.c.i.p. $\square$

No a.c.i.p

2.4 Krengel's null stochastic ergodic theorem

If $\#$ an m -absolutely continuous T -invariant probability, then

$$\frac{1}{n} \sum_{k=0}^{n-1} \widehat{T}^k h \xrightarrow{m} 0 \text{ as } n \rightarrow \infty \quad \forall h \in L^1(m).$$

For the proof of Krengel's theorem, we'll need

2.5 Komlos' theorem

Let $(X, \mathcal{B}, m)$ be a probability space and suppose that

$$f_n \in L^1(m) \quad (n \geq 1), \quad \sup_{n \geq 1} \|f_n\|_1 < \infty,$$

then $\exists f \in L^1(m)$ and $n_k \rightarrow \infty$ such that

$$\frac{1}{K} \sum_{j=1}^K f_{n'_j} \xrightarrow[K \rightarrow \infty]{} f \quad \text{a.e.}$$

whenever $\{n'_j : j \geq 1\}$ is a subsequence of $\{n_k : k \geq 1\}$.

1

Proof of Krengel's theorem

It suffices to prove the theorem for $h \in L^1(m)$, $h \geq 0$, $\int_X h dm > 0$. So let h be such a function and set $A_n := \frac{1}{n} \sum_{k=0}^{n-1} \widehat{T}^k h$, then $\int_X A_n dm = \int_X h dm$ and so by Komlos' theorem for any sequence $m_\ell \rightarrow \infty$, $\exists H \in L^1(m)$ and a subsequence $n_k = m_{\ell_k} \rightarrow \infty$ such that $\frac{1}{N} \sum_{k=0}^{N-1} A_{n_k} \rightarrow H$

1

Proof See

J. Komlós: *A generalization of a problem of Steinhaus* Acta Math. Acad. Sci. Hun. vol.18 pp217–229 (1967) &/or:

M. Schwartz: *New proofs of a theorem of Komlós*; Journal: Acta Math. Hun. Vol. 47, pp 181-185 (1986).

a.e., whence $\widehat{T}H = H$. The absence of an m -absolutely continuous T -invariant probability forces $H = 0$. It follows that $A_{n_k} \rightarrow 0$ in measure. $\square$

Weakly wandering sets.

Let T be a nonsingular transformation of the probability space $(X, \mathcal{B}, m)$. A measurable set $W \in \mathcal{B}$ is called *weakly wandering* if there is a sequence $n_k \rightarrow \infty$ such that $\{T^{-n_k}W : k \geq 1\}$ are disjoint. Evidently, if $\exists$ a T -invariant probability $\mu \sim m$ then there can be no weakly wandering set with positive measure (else $1 \geq \mu(X) = \sum_{k=1}^{\infty} \mu(T^{-n_k}W) = \infty$).

2.6 Theorem (Hajian, Ito, Kakutani)

If $\nexists$ a m -absolutely continuous T -invariant probability, then X is a union of weakly wandering sets mod m .

Proof We'll show that $\forall \epsilon > 0 \exists$ a weakly wandering set $W \in \mathcal{B}$ with $m(W^c) < \epsilon$. Fix $\epsilon > 0$. We first show that $\exists n_k \rightarrow \infty$ and $B_k \in \mathcal{B}$ such that

$$(*) \quad m(B_k^c) < \frac{\epsilon}{2^k} \quad \& \quad \sum_{j=1}^{k-1} m(T^{-(n_k-n_j)} B_k \cap B_j) < \frac{\epsilon}{2^k} \quad \forall k \geq 1.$$

Evidently, $W := \bigcap_{k=1}^{\infty} B_k \setminus \bigcup_{1 \leq i < j < \infty} B_i \cap T^{-(n_j-n_i)} B_j$ is weakly wandering, and it follows from (*) that $m(W) > 1 - 2\epsilon$.

To establish (*) note first that by the null stochastic ergodic theorem, $\forall \epsilon > 0$, $f \in L^1(m)_+$, $\exists B \in \mathcal{B}$ and $n \geq 1$ such that $m(B^c) < \epsilon$ and $\int_X \widehat{T}^n f 1_B dm < \epsilon$.

This enables the choice of $n_k \rightarrow \infty$ and $B_k \in \mathcal{B}$ such that

$$m(B_k^c) < \frac{\epsilon}{2^k} \quad \& \quad \int_{B_k} \widehat{T}^{n_k-n_{k-1}} \left(\sum_{j=1}^{k-1} \widehat{T}^{n_{k-1}-n_j} 1_{B_j} \right) dm < \frac{\epsilon}{2^k} \quad \forall k \geq 1.$$

These satisfy (*). $\square$

RECURRENCE AND CONSERVATIVITY

A set $W \in \mathcal{B}$, $m(W) > 0$ is called *wandering* (for the NPT $(X, \mathcal{B}, m, T)$) if the sets $\{T^{-n}W\}_{n=0}^{\infty}$ are disjoint. and the NPT T is called *conservative* if $\mathcal{W}(T) = \emptyset$ (i.e. there are no wandering sets).

Remarks.

¶1 A conservative NPT $(X, \mathcal{B}, m, T)$ is non-singular. Else $\exists A \in \mathcal{B}$, $m(A) > 0$ with $m(T^{-1}A) = 0$, whence $m(T^{-n}A) = 0 \quad \forall n \geq 1$. It follows that $W := A \setminus \bigcup_{n=1}^{\infty} T^{-n}A$ is a wandering set satisfying $m(W) = m(A)$.

¶2 Similarly, a NPT $(X, \mathcal{B}, m, T)$ is conservative iff (!) it is *incompressible* in the sense that $A \in \mathcal{B}$ and $T^{-1}A \subset A$ imply $A = T^{-1}A \bmod m$.

¶3 If $(X, \mathcal{B}, m, T)$ is a Kakutani skyscraper over the PPT $(\Omega, \mathcal{F}, \mu, S)$, then T is conservative.

Proof of $\Leftarrow$ If T is not conservative, then $\exists A \in \mathcal{F}_+, A \times \{1\} \in \mathcal{W}(T)$ whence $A \in \mathcal{W}(S)$. $\square$

2.7 Halmos recurrence theorem

Let $(X, \mathcal{B}, m, T)$ be a NST. TFAE:

- (i) T is conservative;
- (ii) $A \subset \bigcup_{n=1}^{\infty} T^{-n}A \quad \forall A \in \mathcal{B}_+$;
- (iii) $\sum_{n=1}^{\infty} 1_A \circ T^n = \infty$ a.e. on $A \quad \forall A \in \mathcal{B}_+$.

Proof of (i) $\Rightarrow$ (iii)

Suppose that $A \in \mathcal{B}$, $m(A) > 0$. The set $W := A \setminus \bigcup_{n=1}^{\infty} T^{-n}A$ is wandering if of positive measure, whence $m(W) = 0$ and $A \subseteq \bigcup_{n=1}^{\infty} T^{-n}A \bmod m$. By null preservation, $T^{-N}A \subseteq \bigcup_{n=N+1}^{\infty} T^{-n}A \bmod m \quad \forall N \geq 1$, whence, mod m :

$$A \subseteq \bigcup_{n=1}^{\infty} T^{-n}A \subseteq \dots \subseteq \bigcup_{n=N+1}^{\infty} T^{-n}A \subseteq \dots \subseteq \bigcap_{j=1}^{\infty} \bigcup_{n=j}^{\infty} T^{-n}A = \left[\sum_{n=1}^{\infty} 1_A \circ T^n = \infty \right].$$

$\square$

CONDITIONS FOR CONSERVATIVITY.

2.8 Maharam Recurrence theorem

Let $(X, \mathcal{B}, m, T)$ be MPT.

If $\exists A \in \mathcal{B}$, $m(A) < \infty$ such that $X = \bigcup_{n=1}^{\infty} T^{-n}A \bmod m$, then T is conservative.

Proof We have that $\sum_{n=1}^{\infty} 1_A \circ T^n = \infty$ a.e. If $W \in \mathcal{W}$, $m(W) > 0$, then $\forall n \geq 1$,

$$\begin{aligned} m(A) &\geq \int_{T^{-n}A} \left(\sum_{k=1}^n 1_W \circ T^k \right) dm = \sum_{k=1}^n m(T^{-k}W \cap T^{-n}A) \\ &= \sum_{j=0}^{n-1} m(W \cap T^{-j}A) = \int_W \left(\sum_{j=0}^{n-1} 1_A \circ T^j \right) dm \rightarrow \infty. \end{aligned}$$

Contradiction. $\square$

For example, any PPT is conservative. This statement is known as Poincaré's recurrence theorem.

A MPT of a σ -finite, infinite measure space need not be conservative. For example $x \mapsto x + 1$ is a measure preserving transformation of $\mathbb{R}$ equipped with Borel sets, and Lebesgue measure, which is totally dissipative.

Example.

The original Boole transformation $T : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$T(x) = x - \frac{1}{x}$$

is conservative.

Proof By proposition 2.1, $m \circ T^{-1} = m$. By inspection, $\bigcup_{n=0}^{\infty} T^{-n}[-1, 1] = \mathbb{R}$. $\square$

Exercise 6. Let

$$T(x) = x + \sum_{k=1}^N \frac{p_k}{t_k - x} \text{ where } p_1, \dots, p_N > 0 \text{ \& } t_1, \dots, t_N \in \mathbb{R}.$$

Show that $\bigcup_{n=1}^{\infty} T^{-n}(u, v) = \mathbb{R} \pmod{m}$ where $u := \min T^{-1}\{0\}$ & $v := \max T^{-1}\{0\}$; and hence that T is conservative.

Hint WLOG, $N \geq 2$, $u < 0 < v$ & $T(0) = 0$.

Exercise 7: Skyscraper conservativity.

Let $(X, \mathcal{B}, m, T)$ be a Kakutani skyscraper over the NST $(\Omega, \mathcal{F}, \mu, S)$. Show that T is a NST, and is conservative iff S is conservative.

Exercise 8: Stronger recurrence properties.

Let $(X, \mathcal{B}, m, T)$ be a conservative NST.

Show that if (Y, d) is a separable, metric space and $h : X \rightarrow Y$ is measurable, then

$$\varliminf_{n \rightarrow \infty} d(h, h \circ T^n) = 0 \text{ a.e..}$$

Induced transformation.

This is the “reverse” of the skyscraper construction.

Suppose $(X, \mathcal{B}, m, T)$ is a NST and let $A \in \mathcal{B}_+$ be such that m -a.e. point of A returns to A under iterations of T (e.g. if $(X, \mathcal{B}, m, T)$ is conservative). The *return time* function to A , defined for $x \in A$ by $\varphi_A(x) := \min\{n \geq 1 : T^n x \in A\}$ is finite m -a.e. on A .

The *induced transformation* on A is defined by $T_A x = T^{\varphi_A(x)} x$.

The first key observation is that $(A, \mathcal{B} \cap A, T_A, m_A)$ is a NST and, if T is a MPT, then so is T_A . These follows from

$$T_A^{-1}B = \bigcup_{n=1}^{\infty} [\varphi = n] \cap T^{-n}B.$$

It follows that $\varphi_A \circ T_A$ is defined a.e. on A and an induction now shows that all powers $\{T_A^k\}_{k \in \mathbb{N}}$ are defined a.e. on A , and satisfy

$$T_A^k x = T^{(\varphi_A)_k(x)} x \text{ where } (\varphi_A)_1 = \varphi_A, \quad (\varphi_A)_k = \sum_{j=0}^{k-1} \varphi_A \circ T_A^j.$$

Both constructions can be generalized to the nonsingular case.

Exercise 9: Inducing inverse to skyscraping.

Let $(X, \mathcal{B}, m, T)$ be an invertible, conservative MPT and suppose that $A \in \mathcal{B}$, $m(A) = 1$ satisfies $\bigcup_{n=1}^{\infty} T^{-n} A = X \pmod{m}$.

Show that $(X, \mathcal{B}, m, T)$ is isomorphic to the Kakutani skyscraper over $(A, \mathcal{B} \cap A, m_A, T_A)$ with height function φ_A .

Lecture 3 14/11/2014.**HOPF DECOMPOSITION**

Let $(X, \mathcal{B}, m, T)$ be a NPT. The collection $\mathcal{W}(T)$ of wandering sets is a hereditary collection (any measurable subset of a member is also a member), and T -sub-invariant (W wandering or null $\implies T^{-1}W$ wandering or null).

By **exhaustion**, $\exists$ a countable union of wandering sets $\mathfrak{D}(T) \in \mathcal{B}$ with the property that any wandering set $W \in \mathcal{B}$ is contained in $\mathfrak{D}(T) \bmod m$ (i.e. $m(W \setminus \mathfrak{D}(T)) = 0$). This **measurable union** $\mathfrak{D}(T)$ of $\mathcal{W}(T)$ is unique mod m and $T^{-1}\mathfrak{D} \subseteq \mathfrak{D} \bmod m$. It is called the *dissipative part* of the nonsingular transformation T .

Evidently T is conservative on $\mathfrak{C}(T) := X \setminus \mathfrak{D}(T)$, the *conservative part* of T .

The partition $\{\mathfrak{C}(T), \mathfrak{D}(T)\}$ is called the *Hopf decomposition* of T .

The nonsingular transformation T is called (totally) *dissipative* if $\mathfrak{D}(T) = X \bmod m$.

3.1 Proposition. *Any inner function T with $\alpha > 1$ in $(\clubsuit)$ is dissipative.*

Proof By proposition 2.1,

$$\sum_{n=1}^{\infty} m(T^{-n}A) < \infty \quad \forall A \in \mathcal{B}, 0 < m(A) < \infty$$

and is T is dissipative. $\square$

Exercise 10:

In this exercise, you show that if $(X, \mathcal{B}, m, T)$ is an invertible NST, then $\exists$ a wandering set $W \in \mathcal{B}$ such that

$$\mathfrak{D} = \bigcup_{n \in \mathbb{Z}} T^n W.$$

Hints For $A \in \mathcal{B}$ set $A^T := \bigcup_{n \in \mathbb{Z}} T^n A$.

WLOG, $m(X) = 1$.

- Define $\epsilon_1 := \sup \{m(W) : W \in \mathcal{W}\}$;
- choose $W \in \mathcal{W}$ with $m(W_1) \geq \frac{\epsilon_1}{2}$;
- define $\epsilon_2 := \sup \{m(W) : W \in \mathcal{W}, W \cap W_1^T = \emptyset\}$;
- choose $W_2 \in \mathcal{W}$, $W \cap W_1^T = \emptyset$ with $m(W_2) \geq \frac{\epsilon_2}{2}$.

Continue this process to obtain $\{W_n : n \in \mathbb{N}\} \subset \mathcal{W}$ & $\{\epsilon_n : n \in \mathbb{N}\} \subset \mathbb{R}_+$ so that

- $W_k \cap W_\ell^T = \emptyset \ \forall \ k > \ell$;
- $2m(W_n) \geq \epsilon_n := \sup \{m(W) : W \in \mathcal{W}, W \cap W_k^T = \emptyset \ \forall \ 1 \leq k \leq n-1\}$.

Show that $W := \bigcup_{n \geq 1} W_n$ is as required.

Exercise 11: Hopf decomposition not T -invariant.

Let $(X, \mathcal{B}, m, T) = ([0, 2], \mathcal{B}([0, 2]), \text{Leb})$ where $T : [0, 2) \rightarrow [0, 2)$ is defined by

$$T(x) := \begin{cases} 2x & x \in [0, 1), \\ 1 + (2(x-1) \bmod 1) & x \in [1, 2). \end{cases}$$

Show that T is non-singular, $\mathfrak{C}(T) = [1, 2)$ and that $m(T^{-1}\mathfrak{D}(T)\Delta\mathfrak{D}(T)) = \frac{1}{2}$.

CONSERVATIVITY AND TRANSFER OPERATORS

3.2 Hopf's recurrence theorem

If $T : X \rightarrow X$ is nonsingular and $f \in L^1$, $f > 0$, then

$$\mathfrak{C}(T) = \left[\sum_{n=1}^{\infty} \widehat{T}^n f = \infty \right] \mod m.$$

Proof of $\supseteq$ Fix $W \in \mathcal{W}_T$, then

$$\infty > \int_X f dm \geq \int_X f \left(\sum_{n \geq 0} 1_W \circ T^n \right) dm = \int_W \left(\sum_{n \geq 0} \widehat{T}^n f \right) dm.$$

This shows that $\mathfrak{D}(T) \subset [\sum_{n=1}^{\infty} \widehat{T}^n f < \infty]$. $\square$

Proof of $\subseteq$ Assume otherwise and fix $A \subset \mathfrak{C}(T)$ s.t. $\sum_{n=1}^{\infty} \widehat{T}^n f < \infty$ on A .

WLOG $f(x) \geq c > 0 \ \forall \ x \in A$, and the series converges uniformly on A whence $\int_A (\sum_{n=1}^{\infty} \widehat{T}^n f) dm < \infty$.

On the other hand, by Halmos' recurrence theorem $\sum_{n \geq 0} 1_A \circ T^n = \infty$ a.e. on A .

Thus

$$\begin{aligned} \infty &> \int_A \left(\sum_{n=0}^{\infty} \widehat{T}^n f \right) dm = \int_X f \left(\sum_{n \geq 0} 1_A \circ T^n \right) dm \\ &\geq \int_A f \left(\sum_{n \geq 0} 1_A \circ T^n \right) dm \geq c \int_A \left(\sum_{n \geq 0} 1_A \circ T^n \right) dm = \infty \quad \square \quad \square \end{aligned}$$

Exercise 11: Hopf recurrence theorem for MPTs.

Suppose that T is a MPT of the σ -finite measure space $(X, \mathcal{B}, m)$.

Show that

$$\left[\sum_{n=1}^{\infty} f \circ T^n = \infty \right] = \mathfrak{C}(T) \pmod{m} \quad \forall f \in L^1(m), f > 0.$$

3.3 Corollary *An inner function $T : \mathbb{R} \rightarrow \mathbb{R}$ is conservative iff $\sum_{n=1}^{\infty} \operatorname{Im} \frac{-1}{T^n(\omega)} = \infty$ for some and hence all $\omega \in \mathbb{R}^{2+}$.*

Proof For fixed $\omega \in \mathbb{R}^{2+}$, T is conservative iff $\sum_{n=1}^{\infty} \widehat{T}^n \varphi_{\omega} = \infty$ a.e..
Now

$$\widehat{T}^n \varphi_{\omega}(t) = \widehat{T}^n \varphi_{T^n(\omega)} = \operatorname{Im} \frac{1}{t - T^n(\omega)}$$

and the corollary follows. $\square$

Exercise 12. Let $Tx = x + \beta + \int_{\mathbb{R}} \frac{d\nu(t)}{t-x}$ where ν is a finite, Lebesgue-singular, measure on $\mathbb{R}$ with compact support. Show that T is conservative if $\beta = 0$ and dissipative if $\beta \neq 0$.

Hints Fix $\omega \in \mathbb{R}^{2+}$ and let $T^n \omega = u_n + iv_n$, then

- when $\beta \neq 0$. $\exists B = B(\omega) \in \mathbb{R}_+$ & $C = C(\omega) \in \mathbb{R}$ so that

$$(I) \quad v_n \uparrow B \quad \& \quad u_n = \beta n - \frac{\nu}{\beta} \log n + C + O\left(\frac{\log n}{n}\right) \quad \text{as } n \rightarrow \infty;$$

and

- when $\beta = 0$,

$$(II) \quad \sup_{n \geq 1} |u_n| < \infty \quad \& \quad v_n \sim \sqrt{2\nu n} \quad \text{as } n \rightarrow \infty \quad \text{where } \nu := \nu(\mathbb{R}).$$

ERGODICITY

A transformation T of the measure space $(X, \mathcal{B}, m)$ is called *ergodic* if

$$A \in \mathcal{B}, T^{-1}A = A \pmod{m} \Rightarrow m(A) = 0, \text{ or } m(A^c) = 0.$$

In general, let

$$\mathfrak{I}(T) := \{A \in \mathcal{B}, T^{-1}A = A\}.$$

Remarks.

It is not hard to see that:

- $\mathfrak{I}(T)$ is a σ -algebra (and that T is ergodic iff $\mathfrak{I} \stackrel{m}{=} \{\emptyset, X\}$);
- an invertible ergodic nonsingular transformation of a non-atomic measure space is necessarily conservative;

- a nonsingular transformation $(X, \mathcal{B}, m, T)$ is conservative and ergodic iff

$$\sum_{n=1}^{\infty} 1_A \circ T^n = \infty \text{ a.e. } \forall A \in \mathcal{B}_+.$$

SOME ERGODIC TRANSFORMATIONS

Ergodicity of rank one constructions.

3.4 Proposition

Let $(X, \mathcal{B}, m, T)$ be a rank one MPT as above, then T is ergodic.

Proof Let

$$R_n = \bigcup_{I \in \mathfrak{r}_n} I \uparrow X$$

be the refining sequence of Rokhlin towers defining T ; where each

$$\mathfrak{r}_n = \{T^j I_n : 0 \leq j \leq k_n - 1\}$$

is a partition of R_n into intervals with equal lengths $m(I_n) \xrightarrow{n \rightarrow \infty} 0$.

We claim first that it suffices to show that

For $\epsilon > 0$ & $A \in \mathcal{B}_+$, $\exists N = N_{\epsilon, A}$ so that

$$\forall n > N \exists I \in \mathfrak{r}_n \text{ s.t. } m(A|I) > 1 - \epsilon.$$

Proof of sufficiency of ☞

Suppose that $A \in \mathcal{B}_+$, $TA = A$. We'll show assuming ☞ that $\forall N \geq 1$ large enough,

$$m(A \cap R_N) > (1 - \epsilon)m(R_N) \quad \forall \epsilon > 0$$

whence $A \supset R_N \uparrow X \pmod{m}$.

To see this, choose (by ☞) $n \geq N$ & $J \in \mathfrak{r}_n$ satisfying $m(A|J) > 1 - \epsilon$. Then for each $K = T^{i_K} J \in \mathfrak{r}_n$, we have using T -invariance of m & A :

$$m(A|K) = \frac{m(A \cap T^{i_K} J)}{m(T^{i_K} J)} = m(A|J) > 1 - \epsilon$$

whence

$$m(A \cap R_N) = \sum_{K \in \mathfrak{r}_n, K \subset R_N} m(A|K)m(K) > (1 - \epsilon)m(R_N). \quad \square$$

Proof of ☞

Suppose that $A \in \mathcal{B}_+$ and fix $N \geq 1$ so that $B := A \cap R_N \in \mathcal{B}_+$. For $n \geq N$, let

$$\mathfrak{s}_n := \{I \in \mathfrak{r}_n : I \subset R_N\}.$$

Fix $0 < \epsilon < 1$ and for $n \geq N$ let

$$\mathcal{Z}_n := \{I \in \mathfrak{s}_n : m(B|I) > 1 - \epsilon\} \text{ \& } \mathcal{Y}_n := \mathfrak{s}_n \setminus \mathcal{Z}_n.$$

We show that $\forall n$ large enough, $\mathcal{Z}_n \neq \emptyset$.

Since $\sigma(\bigcup_{n \geq N} \mathfrak{s}_n) = \mathcal{B}(R_N)$, $\exists n \geq N$ & C_n , a union of sets in $\mathfrak{s}_n$ so that $m(B \Delta C_n) < \frac{\epsilon^2 m(B)}{9}$. It follows that

$$\begin{aligned} m(C_n) - \frac{\epsilon^2 m(B)}{9} &< m(B \cap C_n) \\ &= \sum_{I \in \mathfrak{s}_n, I \subset C_n} m(B|I)m(I) \\ &= \sum_{I \in \mathcal{Z}_n, I \subset C_n} m(B|I)m(I) + \sum_{I \in \mathcal{Y}_n, I \subset C_n} m(B|I)m(I) \\ &\leq \sum_{I \in \mathcal{Z}_n, I \subset C_n} m(I) + (1 - \epsilon) \sum_{I \in \mathcal{Y}_n, I \subset C_n} m(I) \\ &= m(\bigcup \mathcal{Z}_n) + (1 - \epsilon)m(C_n) \end{aligned}$$

whence

$$\begin{aligned} m(\bigcup \mathcal{Z}_n) &\geq m(C_n) - \frac{\epsilon^2 m(B)}{9} - (1 - \epsilon)m(C_n) \\ &= \epsilon m(C_n) - \frac{\epsilon^2 m(B)}{9} \\ &> \epsilon m(B) - \frac{\epsilon^3 m(B)}{9} - \frac{\epsilon^2 m(B)}{9} \\ &> \frac{7\epsilon m(B)}{9} > 0. \quad \square \end{aligned}$$

Exercise 13. Here, you show that conservative inner functions are ergodic.

Let T be a conservative inner function and suppose to obtain a contradiction that $A \in \mathcal{B}(\mathbb{R})$, $T^{-1}A = A$, $m(A) > 0$, $m(A^c) > 0$.

Define $v : \mathbb{R}^{2+} \rightarrow [0, \infty)$ by $v(\omega) := P_\omega(A)$.

Show that:

- (i) v is harmonic, not constant and that $v(T(\omega)) = v(\omega) \quad \forall \omega \in \mathbb{R}^{2+}$;
- (ii) there is an analytic function $F : \mathbb{R}^{2+} \rightarrow \mathbb{R}^{2+}$ and $c \in \mathbb{R}$ so that $\text{Im } F = v$ and $F(T(\omega)) = F(\omega) + c$.
- (iii) For $r > 0$ define the bounded analytic $g_r : \mathbb{R}^{2+} \rightarrow \mathbb{C}$ by $g_r(\omega) := e^{irF(\omega)}$. Show that g_r is not constant, and that $\exists r_0 > 0$ with $g_{r_0} \circ T \equiv g_{r_0}$.

(iv) Using Blaschke's theorem (or otherwise) show that g_{r_0} is constant on $\mathbb{R}^{2+}$.

(v) Show that the contradiction implies that T is ergodic.

ERGODICITY VIA STRONGER PROPERTIES

Sometimes it's easier to prove more than ergodicity.

One-sided Bernoulli shifts.

Let $X = \mathbb{R}^{\mathbb{N}}$ and let $\mathcal{B}(X)$ be the σ -algebra generated by *cylinder* sets of form $[A_1, \dots, A_n] := \{\underline{x} \in X : x_j \in A_j, 1 \leq j \leq n\}$, where $A_1, \dots, A_n \in \mathcal{B}(\mathbb{R})$ (the Borel subsets of $\mathbb{R}$), and let the *shift* $S : X \rightarrow X$ be defined by

$$(Sx)_n = x_{n+1}.$$

For $p : \mathcal{B}(\mathbb{R}) \rightarrow [0, 1]$ a probability, let $\mu_p : \mathcal{B}(X) \rightarrow [0, 1]$ be the probability satisfying

$$\mu_p([A_1, \dots, A_n]) = \prod_{k=1}^n p(A_k) \quad (A_1, \dots, A_n \in \mathcal{B}(\mathbb{R})).$$

Evidently $S^{-1}[A_1, \dots, A_n] = [\mathbb{R}, A_1, \dots, A_n]$ whence $\mu_p \circ S^{-1} = \mu_p$.

The *one-sided Bernoulli shift* with marginal distribution p is the probability preserving transformation S of $(X, \mathcal{B}, \mu_p)$.

Tail, exactness. Let T be a nonsingular transformation of $(X, \mathcal{B}, m)$. The *tail* σ -algebra of T is

$$\mathfrak{T}(T) := \bigcap_{n=1}^{\infty} T^{-n}\mathcal{B}.$$

The transformation T is called *exact* if $\mathfrak{T}(T) = \{\emptyset, X\} \bmod m$.

Evidently $\mathfrak{I}(T) \subset \mathfrak{T}(T) \bmod m$ and so exact transformations are ergodic.

Note however, that no invertible nonsingular transformation can be exact (except the identity on a 1-pt. space). Hence an irrational rotation of $\mathbb{T}$ is ergodic, but not exact.

3.5 Kolmogorov's zero-one law

Any one-sided Bernoulli shift is exact.

Proof

Suppose that $B \in \mathcal{B}$ is a finite union of cylinders. If the length of the longest cylinder in the union is n , then

$$\mu_p(B \cap S^{-n}C) = \mu_p(B)\mu_p(C) \quad \forall C \in \mathcal{B}.$$

Now suppose $A \in \mathfrak{T}$. Since, for each $n \in \mathbb{N}$,

$$A = S^{-n}A_n \text{ where } A_n \in \mathcal{B}, \mu_p(A_n) = \mu_p(A),$$

we have that

$$\mu_p(B \cap A) = \mu_p(B)\mu_p(A)$$

for $B \in \mathcal{B}$ a finite union of cylinders, and hence (by approximation) $\forall B \in \mathcal{B}$. This implies that

$$0 = \mu_p(A \cap A^c) = \mu_p(A)(1 - \mu_p(A))$$

demonstrating that $\mathfrak{T}$ is trivial mod μ_p . $\square$

Exercise 14.

Show that a piecewise onto, C^2 , uniformly expanding interval map satisfying Adler's condition is exact.

Hint: For $n \geq 1$, $a \in \alpha_n$: $\widehat{T}^n 1_a = |v'_a| = e^{\pm K} m(a)$.

Nonsingular Adding Machine.

Let $\Omega = \{0, 1\}^{\mathbb{N}}$, and $\mathcal{B}$ be the σ -algebra generated by cylinders. We consider again the *adding machine* $\tau: \Omega \rightarrow \Omega$ defined by

$$\tau(1, \dots, 1, 0, \epsilon_{n+1}, \epsilon_{n+2}, \dots) = (0, \dots, 0, 1, \epsilon_{n+1}, \epsilon_{n+2}, \dots).$$

The adding machine has

the odometer property.

$$\ominus \quad \{((\tau^k x)_1, \dots, (\tau^k x)_n) : 0 \leq k \leq 2^n - 1\} = \{0, 1\}^n \quad \forall x \in \Omega, \quad n \geq 1.$$

The next lemma illustrates how the odometer “parametrizes” the tail of the one-sided shift $S: \Omega \rightarrow \Omega$.

3.6 Lemma

For $x \notin \widetilde{\mathbb{Z}} := \{\tau^n(\bar{0}) : n \in \mathbb{Z}\}$,

$$\{y \in \Omega : \exists n \geq 0, S^n(y) = S^n(x)\} = \{\tau^n(x) : n \in \mathbb{Z}\}.$$

Proof Note that $\widetilde{\mathbb{Z}} = \{x \in \Omega : \exists \lim_{n \rightarrow \infty} x_n\}$. Thus for $x \notin \widetilde{\mathbb{Z}}$, both $\ell(x) := \min\{n \geq 1 : x_n = 0\}$ and $\mathbf{m}(x) := \min\{n \geq 1 : x_n = 1\}$ are finite, whence

$$\exists n \geq 1 \text{ s.t. } S^n x = S^n \tau(x) = S^m \tau^{-1}(x).$$

Since $\tau \widetilde{\mathbb{Z}} = \widetilde{\mathbb{Z}}$,

$$\{y \in \Omega : \exists n \geq 0, S^n(y) = S^n(x)\} \supset \{\tau^n(x) : n \in \mathbb{Z}\}.$$

For the other inclusion, suppose $S^n x = S^n y = z$, then using the odometer property,

$$\underbrace{(0, \dots, 0, z)}_{n \text{ times}} = \tau^{-\nu_n(x)}(x) = \tau^{-\nu_n(y)}(y)$$

where $\nu_n(\omega) := \sum_{k=1}^n 2^{k-1} \omega_n$. Thus

$$y = \tau^{\nu_n(y) - \nu_n(x)}(x). \quad \square$$

For $p \in (0, 1)$, set $\mu_p = \Pi(1-p, p) \in \mathcal{P}(\Omega)$ and recall that

$$\frac{d\mu_p \circ \tau}{d\mu_p} = \left(\frac{1-p}{p} \right)^\phi$$

where $\phi(x) := \min \{n \geq 1 : x_n = 0\} - 2 =: \ell(x) - 2$.

3.7 Proposition

τ is an invertible, conservative, ergodic nonsingular transformation of $(\Omega, \mathcal{B}, \mu_p)$.

Proof It is not hard to show, using lemma 3.6, that $\mathfrak{I}(\tau) = \mathfrak{I}(S) \bmod \mu_p$ and the ergodicity of $(\Omega, \mathcal{B}, \mu_p, \tau)$ follows from the exactness of $(\Omega, \mathcal{B}, \mu_p, S)$. As above, conservativity is automatic in this case. $\square$

3.8 Rigidity proposition For $0 < p < 1$, $(\Omega, \mathcal{B}, \mu_p)$ is rigid in the sense that if $f : \Omega \rightarrow \mathbb{R}$ is measurable, then $\forall \epsilon > 0$,

$$\mu_p([|f \circ \tau^{2^n} - f| \geq \epsilon]) \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Proof Firstly, note that if $f : \Omega \rightarrow \mathbb{R}$ and f is defined by $f(x) = g(x_1, \dots, x_n)$ for some $n \in \mathbb{N}$, then $f \circ \tau^{2^k} \equiv f$ for every $k \geq n$. To enable approximation, we show that $\exists \Delta > 0$ & $M > 1$ so that

$$(\clubsuit) \quad \mu_p(\tau^{-2^n} A) \leq M \mu_p(A)^\Delta \quad \forall A \in \mathcal{B}.$$

Proof of $(\clubsuit)$

As before,

$$\frac{d\mu_p \circ \tau^{-1}}{d\mu_p} = \left(\frac{p}{1-p} \right)^\psi \text{ where } \psi(x) := \min \{n \in \mathbb{N} : x_n = 1\} - 2;$$

Using the odometer property:

$$\begin{aligned} (\spadesuit) \quad \sum_{j=0}^{2^n-1} \psi(\tau^{-j} x) &= \sum_{\epsilon \in \{0,1\}^n \setminus \{1\}} \psi(\epsilon) + n + \psi(S^n x) \\ &= \sum_{k=1}^n (k-2)2^{n-k} + n + \psi(S^n x) \\ &= \psi(S^n x). \end{aligned}$$

By (☕)

$$\begin{aligned} \frac{d\mu_p \circ \tau^{-2^n}}{d\mu_p} &= \prod_{k=0}^{2^n-1} \left(\frac{d\mu_p \circ \tau^{-1}}{d\mu_p} \right) \circ \tau^{-k} \\ &= \prod_{k=0}^{2^n-1} \left(\frac{p}{1-p} \right)^{\psi \circ \tau^{-k}} \\ &= \left(\frac{p}{1-p} \right)^{\psi \circ S^n}. \end{aligned}$$

Fix (!) $q > 1$ be such that $\frac{p^q}{(1-p)^{q-1}} < 1$, then

$$M^q := \left\| \left(\frac{p}{1-p} \right)^{\psi} \right\|_{L^q(\mu_p)}^q \propto \sum_{n \geq 1} \left(\frac{p^q}{(1-p)^{q-1}} \right)^n < \infty$$

and for $A \in \mathcal{B}$,

$$\mu_p(\tau^{-2^n} A) = \int_A \left(\frac{p}{1-p} \right)^{\psi \circ S^n} d\mu_p \leq \left\| \left(\frac{p}{1-p} \right)^{\psi} \right\|_q \mu_p(A)^{\frac{q-1}{q}} = M \mu_p(A)^{\frac{q-1}{q}}$$

by Hölder's inequality. $\square(\otimes)$

Now, suppose that $F : \Omega \rightarrow \mathbb{R}$ is measurable, and let $\epsilon > 0$ be given. There exist $n \in \mathbb{N}$, and $f : \Omega \rightarrow \mathbb{R}$ and f defined by $f(x) = g(x_1, \dots, x_n)$ for some $g : \{0, 1\}^n \rightarrow \mathbb{R}$ such that $\mu_p([|F - f| \geq \epsilon/2]) < \epsilon$. For $k \geq n$, we have $f \circ \tau^{2^k} \equiv f$, whence

$$\begin{aligned} \mu_p([|F \circ \tau^{2^k} - F| \geq \epsilon]) &\leq \mu_p([|F \circ \tau^{2^k} - f \circ \tau^{2^k}| \geq \epsilon/2]) + \mu_p([|F - f| \geq \epsilon/2]) \\ &\leq \epsilon + M \epsilon^{\frac{1}{q}}, \end{aligned}$$

establishing that indeed

$$F \circ \tau^{2^n} \xrightarrow{\mu_p} F. \quad \square$$

Ergodic Maharam extension for the non-singular adding machine.

Define $\tau_\phi : \Omega \times \mathbb{Z} \rightarrow \Omega \times \mathbb{Z}$ by

$$\tau_\phi(x, z) := (\tau x, z + \phi(x)).$$

For $0 < p < 1$ define the measure $m_p : \mathcal{B}(\Omega \times \mathbb{Z}) \rightarrow [0, \infty]$ by

$$m_p(A \times \{z\}) := \mu_p(A) \left(\frac{p}{1-p} \right)^z.$$

This kind of transformation is aka a **discrete Maharam extension**.

3.9 Theorem *For each $0 < p < 1$, $(\Omega \times \mathbb{Z}, \mathcal{B}(\Omega \times \mathbb{Z}), m_p, \tau_\phi)$ is a conservative, ergodic measure preserving transformation.*

Proof that $m_p \circ \tau_\phi = m_p$

Any $A \in \mathcal{B}(\Omega \times \mathbb{Z})$ has a measurable decomposition $A = \bigcup_{z, \ell \in \mathbb{Z}} A_{z, \ell} \times \{z\}$ where $\phi = \ell$ on $A_{z, \ell}$. Thus:

$$\begin{aligned} m_p(\tau_\phi A) &= \sum_{z, \ell \in \mathbb{Z}} m_p(\tau_\phi(A_{z, \ell} \times \{z\})) = \sum_{z, \ell \in \mathbb{Z}} m_p(\tau A_{z, \ell} \times \{z + \ell\}) \\ &= \sum_{z, \ell \in \mathbb{Z}} \mu_p(\tau A_{z, \ell}) \left(\frac{p}{1-p}\right)^{z+\ell} = \sum_{z, \ell \in \mathbb{Z}} \mu_p(A_{z, \ell}) \left(\frac{p}{1-p}\right)^z \\ &= \sum_{z, \ell \in \mathbb{Z}} m_p((A_{z, \ell} \times \{z\})) = m_p(A). \quad \square \end{aligned}$$

Proof of ergodicity of τ_ϕ Suppose that $F : \Omega \times \mathbb{Z} \rightarrow \mathbb{R}$ is bounded, measurable and τ_ϕ -invariant. We'll show first that $F(x, z) = F(x, z-1)$ m_p -a.e..

A similar calculation to $(\clubsuit)$ shows that

$$(\clubsuit) \quad \phi_{2^n}(x) = \phi(S^n x).$$

Iterating τ_ϕ , we have that

$$F(x, z) = F \circ \tau^{2^n}(x, z) = F(\tau^{2^n} x, z + \phi_{2^n}(x)) = F(\tau^{2^n} x, z + \phi(S^n(x))).$$

By the rigidity proposition, $\exists n_k \rightarrow \infty$ and $\Omega_0 \in \mathcal{B}(\Omega)$, $\mu_p(\Omega_0) = 1$ such that

$$F(\tau^{2^{n_k}} x, z) \xrightarrow[k \rightarrow \infty]{} F(x, z) \quad \forall x \in \Omega_0, z \in \mathbb{Z}.$$

The events

$$A_n = [\phi \circ S^n = -1] = \{x \in \Omega : x_{n+1} = 0\}$$

are independent under μ_p , and $\mu_p(A_n) = 1 - p$.

By the Borel-Cantelli lemma, $\exists \Omega_1 \in \mathcal{B}(\Omega)$, $\Omega_1 \subset \Omega_0$, $\mu_p(\Omega_1) = 1$ such that $\forall x \in \Omega_1$, $\exists k_\ell = k_\ell(x) \rightarrow \infty$ with

$$\phi(S^{k_\ell} x) = -1 \quad \forall \ell \geq 1,$$

whence

$$F(x, z) = F(\tau^{2^{k_\ell}} x, z + \phi(S^{k_\ell}(x))) = F(\tau^{2^{k_\ell}} x, z - 1) \xrightarrow[\ell \rightarrow \infty]{} F(x, z - 1).$$

Thus $\exists f : \Omega \rightarrow \mathbb{R}$, measurable, such that $F(x, z) = f(x)$ μ_p -a.e. $\forall z \in \mathbb{Z}$. Since F is τ_ϕ -invariant, f is τ -invariant and μ_p -a.e. constant by ergodicity of $(\Omega, \mathcal{B}, \mu_p, \tau)$. $\square$

Exercise 15.

Show that when $p \neq \frac{1}{2}$, the nonsingular adding machine $(\Omega, \mathcal{B}, \mu_p, \tau)$ has no σ -finite, absolutely continuous, invariant measure.

Hints: Suppose otherwise, that $m \ll \mu_p$ is a σ -finite, τ -invariant measure and let $dm = h d\mu_p$ where $h \geq 0$ is measurable. Show that

- (i) $h > 0$ μ_p -a.e. and $\tau' = \frac{h}{h \circ \tau}$.
- (ii) $\phi = k - k \circ \tau$ where $k : \Omega \rightarrow \mathbb{R}$ satisfies $h = (\frac{1-p}{p})^k$.
- (iii) $F : \Omega \times \mathbb{Z} \rightarrow \mathbb{R}$ defined by $F(x, z) = z + k(x)$ is a non-constant, τ_ϕ -invariant measurable function.....

Lecture 4 16/11/2014.**RATIO ERGODIC THEOREM**

Suppose that $(X, \mathcal{B}, m, T)$ is a conservative, nonsingular transformation.

4.1 Hurewicz's Ergodic Theorem

$$\frac{\sum_{k=1}^n \widehat{T}^k f(x)}{\sum_{k=1}^n \widehat{T}^k p(x)} \xrightarrow{n \rightarrow \infty} E_{m_p} \left(\frac{f}{p} \middle| \mathfrak{I} \right) (x) \text{ for a.e. } x \in X, \forall f, p \in L^1(m), p > 0,$$

where $dm_p = p dm$, and $\mathfrak{I}$ is the σ -algebra of T -invariant sets in $\mathcal{B}$.

Conditional expectations.

Here, given a probability space $(\Omega, \mathcal{F}, P)$, and a sub- σ -algebra $\mathcal{C} \subset \mathcal{F}$, the *conditional expectation wrt $\mathcal{C}$* is a linear operator

$$f \mapsto E_P(f|\mathcal{C}), \quad L^1(\Omega, \mathcal{F}, P) \rightarrow L^1(\Omega, \mathcal{C}, P)$$

satisfying

$$\int_C E_P(f|\mathcal{C}) dP = \int_C f dP \quad \forall C \in \mathcal{C}.$$

Such operators are unique by their defining equations,. They exist $L^2(\Omega, \mathcal{F}, P) \rightarrow L^2(\Omega, \mathcal{C}, P)$ as **orthogonal projections** and extend to L^1 by approximation.

In case T is ergodic,

$$E_{m_p} \left(\frac{f}{p} \middle| \mathfrak{I} \right) = \int_X \frac{f}{p} dm_p = \int_X f dm.$$

Proof of Hurewicz's theorem

Set, for $f, p \in L^1(m)$, $p > 0$, $\widehat{S}_0 f = 0$, and $n \in \mathbb{N}$,

$$\widehat{S}_n f := \sum_{k=0}^{n-1} \widehat{T}^k f, \quad R_n(f, p) := \frac{\widehat{S}_n f}{\widehat{S}_n p}.$$

Let

$$\mathcal{H}_p := \{f = hp + g - \widehat{T}g \in L^1(m) : h \circ T = h \in L^\infty(m), g \in L^1(m)\}.$$

We claim that for $f = hp + g - \widehat{T}g \in \mathcal{H}_p$,

$$R_n(f, p) = h + \frac{g - \widehat{T}^n g}{\widehat{S}_n p}.$$

We show that $R_n(hp, p) = h$ where $h \circ T = h \in L^\infty(m)$. For $g \in L^\infty(m)$, $n \in \mathbb{N}$,

$$\int_X \widehat{T}^n(hp) \cdot g dm = \int_X phg \circ T^n dm = \int_X ph \circ T^n g \circ T^n dm = \int_X h \widehat{T}^n p \cdot g dm$$

for every whence $\widehat{T}^n f = h \widehat{T}^n p$, and $R_n(f, p) = h$. The convergence

$$R_n(f, p) \xrightarrow[n \rightarrow \infty]{} h, \text{ a.e. } \forall f = hp + g - \widehat{T}g \in \mathcal{H}_p$$

follows immediately from the

4.2 Chacon-Ornstein Lemma

$$\frac{\widehat{T}^n g}{\widehat{S}_n p} \xrightarrow[n \rightarrow \infty]{} 0, \text{ a.e. } \forall g \in L^1(m).$$

Proof Choose $\epsilon > 0$, and let $\eta_n = 1_{[\widehat{T}^n g > \epsilon \widehat{S}_n p]}$. We must show that $\sum_{n=1}^\infty \eta_n < \infty$ a.e. $\forall \epsilon > 0$.

We have

$$\epsilon p + \widehat{T}^{n+1} g - \epsilon \widehat{S}_{n+1} p = \widehat{T}(\widehat{T}^n g - \epsilon \widehat{S}_n p),$$

whence

$$\epsilon p + \widehat{T}^{n+1} g - \epsilon \widehat{S}_{n+1} p \leq \widehat{T}(\widehat{T}^n g - \epsilon \widehat{S}_n p)_+,$$

where g_+ denotes $g \vee 0$, $f \vee g = \max\{f, g\}$.

Multiplying both sides of the inequality by η_{n+1} :

$$\begin{aligned} \eta_{n+1} \epsilon p + \eta_{n+1} (\widehat{T}^{n+1} g - \epsilon \widehat{S}_{n+1} p) &= \eta_{n+1} \epsilon p + (\widehat{T}^{n+1} g - \epsilon \widehat{S}_{n+1} p)_+ \\ &\leq \eta_{n+1} \widehat{T}(\widehat{T}^n g - \epsilon \widehat{S}_n p)_+ \\ &\leq \widehat{T}(\widehat{T}^n g - \epsilon \widehat{S}_n p)_+. \end{aligned}$$

Equivalently,

$$\eta_{n+1} \epsilon p \leq \widehat{T} J_n - J_{n+1}$$

where $J_n := (\widehat{T}^n g - \epsilon \widehat{S}_n p)_+$.

Integrating, we get

$$\epsilon \int_X p \eta_{n+1} dm \leq \int_X (J_n - J_{n+1}) dm$$

and, summing over n , we get

$$\epsilon \int_X p \sum_{n=2}^N \eta_n dm \leq \int_X J_1 dm < \infty.$$

This shows that indeed

$$\sum_{n=1}^\infty \eta_n < \infty \text{ a.e.}$$

and thereby proves the lemma. $\square$

We next establish that

$$\textcircled{\smile} \quad \overline{\mathcal{H}_p} = L^1(m).$$

To see this, we show that

$$k \in L^\infty(m), \int_X k f dm = 0 \quad \forall f \in \mathcal{H}_p \quad \Rightarrow \quad k = 0 \quad \text{a.e.}$$

To see this, let

$$k \in L^\infty(m) \quad \ni \quad \int_X k f dm = 0 \quad \forall f \in \mathcal{H}_p,$$

then, in particular

$$\int_X g k \circ T dm = \int_X \widehat{T}g \cdot k dm = \int_X g k dm \quad \forall g \in L^1(m),$$

whence $k \circ T = k$ a.e., and $k p \in \mathcal{H}_p$.

Hence,

$$\int_X k^2 p dm = 0 \quad \Rightarrow \quad k = 0 \quad \text{a.e.}$$

$\textcircled{\smile}$ now follows from the Hahn-Banach theorem. $\square$

Proof of Hurewicz's theorem ctd.

Identification of the limit.

We now identify the limit of $R_n(f, p)$ $f \in \mathcal{H}_p$. Define $\Phi_p : L^1(m) \rightarrow L^1(m_p)$ by

$$\Phi_p(f) := E_{m_p}\left(\frac{f}{p} \parallel \mathfrak{I}\right),$$

then

$$\|\Phi_p(f)\|_{L^1(m_p)} \leq \|f\|_1 \quad \forall f \in L^1(m).$$

We claim that

$$(\spadesuit) \quad R_n(f, p) \xrightarrow{n \rightarrow \infty} \Phi_p(f) \quad \forall f \in \mathcal{H}_p.$$

For this, it suffices that

$$\Phi_p(hp + g - \widehat{T}g) = h \quad \forall f = hp + g - \widehat{T}g \in \mathcal{H}_p.$$

Indeed, if $k \circ T = k \in L^\infty(m)$, then

$$\begin{aligned} \int_X k \frac{f}{p} dm_p &= \int_X k f dm \\ &= \int_X k(hp + g - \widehat{T}g) dm \\ &= \int_X k h p dm + \int_X k(g - \widehat{T}g) dm \\ &= \int_X k h dm_p. \quad \square \end{aligned}$$

We extend $(\star)$ to all $f \in L^1(m)$, by an approximation argument which uses the

4.3 Maximal inequality

For $f, p \in L^1$, such that $p > 0$ a.e., and $t \in \mathbb{R}_+$,

$$m_p([\sup_{n \in \mathbb{N}} R_n(f, p) > t]) \leq \frac{\|f\|_1}{t},$$

where $dm_p = p dm$.

Proof of theorem 4.1 given the maximal inequality

Let $f \in L^1(m)$. Fix $\epsilon > 0$.

By $\odot$, we can write $f = g + k$, where $g \in \mathcal{H}_p$ and $\|k\|_1 < \epsilon^2$. It follows that

$$\overline{\lim}_{n \rightarrow \infty} |R_n(f, p) - \Phi_p(f)| \leq \sup_{n \in \mathbb{N}} |R_n(k, p)| + |\Phi_p(k)|,$$

whence, by the maximal inequality, and by Tchebychev's inequality,

$$\begin{aligned} m_p([\overline{\lim}_{n \rightarrow \infty} |R_n(f, p) - \Phi_p(f)| > 2\epsilon]) &\leq m_p([\sup_{n \geq 1} |R_n(k, p)| > \epsilon]) + m_p([|\Phi_p(k)| > \epsilon]) \\ &\leq \frac{2\|k\|_1}{\epsilon} \leq 2\epsilon. \end{aligned}$$

This last inequality holds for arbitrary $\epsilon > 0$, whence

$$\overline{\lim}_{n \rightarrow \infty} |R_n(f, p) - \Phi_p(f)| = 0 \text{ a.e.,}$$

and the ergodic theorem is almost established, it remaining only to prove the maximal inequality.

4.4 Hopf's Maximal ergodic theorem

$$\int_{[M_n f > 0]} f dm \geq 0, \quad \forall f \in L^1(m), n \in \mathbb{N},$$

where

$$M_n f = \left(\bigvee_{k=1}^n \widehat{S}_k f \right)_+ = \left(\bigvee_{k=0}^n \widehat{S}_k f \right).$$

Proof Note first that if $M_n f(x) > 0$, then

$$\begin{aligned} M_n f(x) &\leq M_{n+1} f(x) = \bigvee_{k=1}^{n+1} \widehat{S}_k f(x) \\ &= f(x) + \bigvee_{k=0}^n \widehat{S}_k \widehat{T} f(x) = f(x) + M_n \widehat{T} f(x). \end{aligned}$$

Also (!) $M_n \widehat{T} f \leq \widehat{T} M_n f$, whence

$$M_n f > 0 \Rightarrow f \geq M_n f - \widehat{T} M_n f,$$

and

$$\int_{[M_n f > 0]} f dm \geq \int_{[M_n f > 0]} (M_n f - \widehat{T} M_n f) dm.$$

Since $\widehat{T} M_n f \geq 0$ a.e., and $M_n f = 0$ on $[M_n f > 0]^c$, we get

$$\begin{aligned} \int_{[M_n f > 0]} f dm &\geq \int_{[M_n f > 0]} M_n f dm - \int_{[M_n f > 0]} \widehat{T} M_n f dm \\ &\geq \int_X M_n f dm - \int_X \widehat{T} M_n f dm \\ &= 0, \end{aligned}$$

whence the theorem. $\square$

Proof of the maximal inequality Suppose f, p, t are as in the maximal inequality, then

$$M_n(f - tp) > 0 \Leftrightarrow \max_{1 \leq k \leq n} R_k(f, p) > t.$$

Thus, using Hopf's maximal ergodic theorem, we obtain

$$\int_{[M_n(f - tp) > 0]} (f - tp) dm \geq 0,$$

whence

$$\begin{aligned} tm_p([\max_{1 \leq k \leq n} R_k(f, p) > t]) &\leq \int_{[\max_{1 \leq k \leq n} R_k(f, p) > t]} f dm \\ &\leq \|f\|_1. \end{aligned}$$

The maximal inequality follows from this as $n \rightarrow \infty$. $\square$

Hurewicz's ergodic theorem is now established.

Hurewicz's theorem for a conservative, ergodic nonsingular transformation T , states that

$$\frac{\sum_{k=0}^{n-1} \widehat{T}^k f(x)}{\sum_{k=0}^{n-1} \widehat{T}^k g(x)} \rightarrow \frac{\int_X f dm}{\int_X g dm} \text{ for a.e. } x \in X$$

whenever $f, g \in L^1(m)$, $\int_X g dm \neq 0$.

Exercise 17: von Neumann's ergodic theorem.

Let $\mathcal{H}$ be a Hilbert space and let $U : \mathcal{H} \rightarrow \mathcal{H}$ be a unitary operator.

Show that

(i) $\mathcal{H}_0 := \{f \in \mathcal{H} : Uf = f\}$ is a closed, invariant subspace of $\mathcal{H}$ and that

$$(ii) \quad \left\| \frac{1}{n} \sum_{k=0}^{n-1} U^k f - Pf \right\| \xrightarrow{n \rightarrow \infty} 0 \quad \forall f \in \mathcal{H}$$

where $P : \mathcal{H} \rightarrow \mathcal{H}_0$ is orthogonal projection.

Exercise 18: Hopf's ergodic theorem.

Suppose that $(X, \mathcal{B}, m, T)$ is a conservative measure preserving transformation.

(i) Prove that

$$\frac{\sum_{k=1}^n f(T^k x)}{\sum_{k=1}^n p(T^k x)} \xrightarrow{n \rightarrow \infty} E_{m_p}(f|\mathfrak{I})(x) \text{ for a.e. } x \in X, \forall f, p \in L^1(m), p > 0.$$

Hint Hopf's ergodic theorem is a special case of Hurewicz's theorem in case T is invertible. It can be proved analogously for T non-invertible.

(ii) Now suppose that T is a conservative, ergodic, measure preserving transformation of the σ -finite, infinite measure space $(X, \mathcal{B}, m)$. Prove that

$$\frac{1}{n} \sum_{k=1}^n f(T^k x) \xrightarrow{n \rightarrow \infty} 0 \text{ for a.e. } x \in X, \forall f \in L^1(m).$$

Pointwise dual ergodicity.

A conservative, ergodic, measure preserving transformation $(X, \mathcal{B}, m, T)$ is called *pointwise dual ergodic* if there are constants $a(n)$ such that

$$(\mathfrak{A}) \quad \frac{1}{a(n)} \sum_{k=0}^{n-1} \widehat{T}^k f \rightarrow \int_X f dm_T \text{ a.e. as } n \rightarrow \infty \forall f \in L^1(X_T).$$

4.5 Proposition Let $Tx = x + \int_{\mathbb{R}} \frac{d\nu(t)}{t-x}$ where ν is a finite, Lebesgue-singular, measure on $\mathbb{R}$ with compact support, then T is pointwise dual ergodic with normalizing constants

$$a(n) = \frac{1}{\pi} \sqrt{\frac{2n}{\nu(\mathbb{R})}}.$$

Proof It follows from the hints to exercise 12 that

$$\frac{1}{a(n)} \widehat{S}_n \varphi_\omega(t) \xrightarrow{n \rightarrow \infty} 1 \quad \forall \omega \in \mathbb{R}^{2+}, t \in \mathbb{R}$$

where $\widehat{S}_n p := \sum_{k=0}^{n-1} \widehat{T}^k p$.

Set $p := \varphi_i$, then by Hurewicz's theorem, for $f \in L^1(m)$ and a.e. $x \in X$,

$$\frac{\widehat{S}_n f(x)}{a(n)} \sim \frac{\widehat{S}_n f(x)}{\widehat{S}_n p(x)} \xrightarrow{n \rightarrow \infty} E_{m_p}(f|\mathfrak{I}).$$

On the other hand, for $f = g * \varphi_{ib}$, $g \in C_C(\mathbb{R})$,

$$f(x) := \int_{\mathbb{R}} g(t) \varphi_{ib}(x-t) dt = \int_{\mathbb{R}} g(t) \varphi_{t+ib}(x) dt$$

whence

$$\widehat{T}^n f = \int_{\mathbb{R}} g(t) \varphi_{T^n(t+ib)}(x) dt$$

and by (I)

$$\frac{\widehat{S}_n f(x)}{a(n)} = \int_{\mathbb{R}} g(t) \frac{\widehat{S}_n \varphi_{t+ib}}{a(n)} dt \xrightarrow{n \rightarrow \infty} \int_{\mathbb{R}} g dm = \int_{\mathbb{R}} f dm$$

whence $E_{m_p}(f|\mathfrak{I})$ is constant.

Since such f are dense in $L^1(m)$, T is ergodic,

$$E_{m_p}\left(\frac{f}{p}|\mathfrak{I}\right) = \int_X f dm \quad \forall f \in L^1$$

and T is pointwise dual ergodic. $\square$

More generally

4.6 Proposition *A conservative inner function $T : \mathbb{R}^{2+} \rightarrow \mathbb{R}^{2+}$ is pointwise dual ergodic.*

Proof

Fix $\omega \in \mathbb{R}^{2+}$ and write $T^n(\omega) := u_n + iv_n$, then by conservativity and corollary 3.3:

$$a(n) := \frac{1}{\pi} \sum_{k=0}^{n-1} \operatorname{Im} \frac{-1}{T^k(\omega)} \rightarrow \infty$$

and

$$\frac{1}{a(n)} \sum_{k=0}^{n-1} \widehat{T}^k \varphi_{\omega} \xrightarrow{n \rightarrow \infty} 1 \quad \forall x \in \mathbb{R}.$$

By exercise 13, T is ergodic whence, by Hurewicz's theorem, for $f \in L^1(m)$,

$$\frac{\widehat{S}_n f(x)}{a(n)} \sim \frac{\widehat{S}_n f(x)}{\widehat{S}_n \varphi_{\omega}(x)} \xrightarrow{n \rightarrow \infty} \int_{\mathbb{R}} f dm \quad \text{a.e.} \quad \square$$

Exercise 19.

Show that Glaisher's transformation $Tx := x + \tan x$ is a conservative inner function with $a(n) \propto \log n$.

Exercise 20.

Here you show that $x \mapsto T(x) := \{\frac{1}{1-x}\} [0, 1] \rightarrow [0, 1]$ is conservative and ergodic but with no a.c.i.p..

(i) Define the measure μ on $[0, 1]$ by $d\mu(x) = \frac{dx}{x}$. Show that $\mu \circ T^{-1} = \mu$.

(ii) For $0 \leq y < \frac{1}{2}$, let $\nu(y) := \min \{k \geq 1 : T^k(y) \geq \frac{1}{2}\}$.

Show that for $y \in (0, \frac{1}{2})$, $k \leq \nu(y)$, $T^k(y) = \frac{y}{1-ky}$, and that $\nu(y) = \kappa$ for $\frac{1}{\kappa+2} \leq y < \frac{1}{\kappa+1}$.

(iii) Show that $\forall x \in [\frac{1}{2}, 1) \exists n \geq 1$ so that $T^n(x) \in [\frac{1}{2}, 1)$.

(iv) Show that the induced transformation $S = T|_{[\frac{1}{2}, 1)} : [\frac{1}{2}, 1) \rightarrow [\frac{1}{2}, 1)$ is a piecewise onto, uniformly expanding, C^2 interval map satisfying Adler's condition.

(v) Using exercise 14 (or otherwise), show that T is conservative and ergodic.

(vi) Using the ergodic theorems (or otherwise) show that T has no a.c.i.p..

Exercise 21: Dissipative exact MPTs.

Let $\Omega = \{0, 1\}^{\mathbb{N}}$ let $S : \Omega \rightarrow \Omega$ be the shift, let $\tau : \Omega \rightarrow \Omega$ be the adding machine and let $\mu_p = \prod(1-p, p) \in \mathcal{P}(\Omega)$, $(0 < p < 1)$. Define $f, \phi : \Omega \rightarrow \mathbb{Z}$ by

$$f(x) := x_1 \quad \& \quad \phi(x) := \ell(x) - 2, \quad \ell(x) := \min \{n \geq 1 : x_n = 0\}$$

and S_f, τ_ϕ by

$$S_f(x, z) = (\sigma(x), z + x_1), \quad \tau_\phi(x, z) := (\tau(x), z + \ell(x) - 2).$$

Show that

(i) $(\Omega \times \mathbb{Z}, \mathcal{B}(\Omega \times \mathbb{Z}), \mu_p \times \#, S_f)$ is a totally dissipative MPT;

(ii) $\mathfrak{T}(S_f) = \mathfrak{T}(\tau_\phi)$.

(iii) $(\Omega \times \mathbb{Z}, \mathcal{B}(\Omega \times \mathbb{Z}), \mu_p \times \#, S_f)$ is exact.

End of lecture notes